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TECHNICAL MEMORANDUM: CALCULATION PROCEDURES AND SPECIFICATION CRITERIA FOR LIGHTING CALCULATIONS

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Prepared for the IES by the
Measurements and Calculations Subcommittee
of the IES Roadway Lighting Committee

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CONTENTS

1.0 Introduction and Scope

This Technical Memorandum establishes consistent methods for:

- Specification of criteria
- Precision of calculations
- Comparisons of calculations to criteria

2.0 Recommendations for Performing Calculations and Making Comparisons

2.1 Precision in Calculations

Calculations shall be carried out to the greatest appropriate precision, and intermediate values shall not be rounded.

2.2 Comparison against a Criterion

When calculated values are to be evaluated against a criterion, the values shall be rounded before comparison to the same precision as the criterion.

For example, if the criterion is presented as 8, then a calculated value of 7.7283 is to be rounded to 8 before comparison. However, if the criterion is presented as 8.0, then the calculated value is rounded to 7.7 before comparison.

2.3 Rounding Technique

Rounding shall conform to the mathematical technique wherein values of less than half the unit value of the last digit position after rounding shall not be increased and values of half or greater shall be increased.

3.0 Recommendations for Specifying Criteria

3.1 Explicit and Implicit Statements of Precision

The precision of a criterion may be explicitly stated so that there is no question of intended tolerance. If the precision of a criterion is not explicitly stated, then the precision is implicitly interpreted as plus or minus half the order of magnitude of the last digit shown. For a criterion expressed as an integer value, a decimal point is assumed after the last digit. **Table 1** provides several examples.

Table 1: Examples

Criterion	Precision
1×10^5	$\pm 0.5 \times 10^5$
1,000 ±50	±50
1,000	±0.5
120	±0.5
110 ±5	±5
110.	±0.5
50	±0.5
5	±0.5
1	±0.5
1.0	±0.05
1.00	±0.005
0.25	±0.005

3.2 Consistency

The precision of criteria shall be consistent among related criteria.

3.3 Significant Digits

In general, criteria and calculations should not be expected to have more than three significant digits.

3.4 Ratios

Ratios should be presented as values with respect to 1; e.g., 6:1, 6.0:1. The explicit or implicit precision is associated with the first value.

4.0 Applicability of Guidelines

The guidelines in this Technical Memorandum apply to criteria set after the issuance of this Standard. They do not apply retroactively.

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